LearnCarbon

A tool for machine learning prediction of global warming potential from abstract designs

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The new construction that is projected to take place between 2020 and 2040 plays a critical role in embodied carbon emissions. The change in material selection is inversely proportional to the budget as the project progresses. Given the fact that early-stage design processes often do not include environmental performance metrics, there is an opportunity to investigate a toolset that enables early-stage design processes to integrate this type of analysis into the preferred workflow of concept designers. The value here is that early-stage environmental feedback can inform the crucial decisions that are made in the beginning, giving a greater chance for a building with better environmental performance in terms of its life cycle. This paper presents the development of a tool called LearnCarbon, as a plugin of Rhino3d, used to educate architects and engineers in the early stages about the environmental impact of their design. It facilitates two neural networks trained with the Embodied Carbon Benchmark Study by Carbon Leadership Forum, which learns the relationship between building geometry, typology, and construction type with the Global Warming potential (GWP) in tons of C02 equivalent (tCO2e). The first one, a regression model, can predict the GWP based on the massing model of a building, along with information about typology and location. The second one, a classification model, predicts the construction type given a massing model and target GWP. LearnCarbon can help improve the building life cycle impact significantly through early predictions of the structure's material and can be used as a tool for facilitating sustainable discussions between the architect and the client.

Keywords: *Machine Learning, Carbon Emissions, LCA, Rhino Plug-In.*

INTRODUCTION

Embodied emissions (EE) play a critical role in the total global emissions due to the new construction that shall take place until 2040 (Architecture 2030, 2018). According to Architecture 2030, the EE from all buildings, infrastructure, and associated materials shall meet a maximum Global Warming Potential (GWP) of 40% below the industry average today and 65% or better in 2030. Additionally, studies have shown that structure is the highest contributor (50-80%) in the Whole Building Life Cycle Assessment (WBLCA) (Embodied Carbon in Building Materials for Real Estate, n.d.), which is the focus of this paper.

Early assessment of EE can save more carbon in the building stock. Further, the cost of making any change in a construction project is directly proportional to the stage of the project (Holzer, 2011). This implies that the price to make changes is higher in the detailed design phase. Currently, most Life Cycle Assessment (LCA) and carbon assessment tools do not address the early design stages and rely on a detailed proposal. As a result, this analysis is undertaken when most decisions have been made, making it very difficult to propose changes.

Through this paper, we present a digital tool called LearnCarbon designed to predict EE from a building envelope. Additionally, the tool predicts the structural inclination to achieve an EE target. The primary objective is to create an integrated tool for the current architectural workflow. This can help incorporate EE as a factor in early-stage design. It is designed as a plugin for Rhino3D (Rhinoceros 3D, 2019) to enable direct implementation in the conceptual sketching phase. LearnCarbon is a pilot project developed as part of the AEC Hackathon 2021 (BLOXHUB, n.d.), held in Copenhagen, Denmark, and it acts as a proof of concept for using Machine Learning (ML) for predicting GWP.

BACKGROUND

There has been significant development in the tools that address LCA in different scopes and stages of construction. One-Click LCA's Carbon Designer (oneclicklca.zendesk.com, 2022) uses a reference basis to calculate the early-stage tonnes of carbon dioxide equivalent (tCO2e) in buildings. (According to Eurostat statistics, A carbon dioxide equivalent or CO2 equivalent, abbreviated as CO2-eq is a metric measure used to compare the emissions from various greenhouse gases on the basis of their global-warming potential (GWP), by converting amounts of other gases to the equivalent amount of carbon dioxide with the same global warming potential. Carbon dioxide equivalents are expressed as tonnes of carbon dioxide equivalents, abbreviated as tCO2e.) Other tools using an analytical model and built as a plugin for Grasshopper, a graphical algorithm editor integrated with Rhino3D, Cardinal LCA (Cardinal LCA, n.d.), CAALA (Food4Rhino, 2021), and Bombyx (Food4Rhino, 2018). Tools built as plugins for later design stages include Tally (Tally, n.d.) and One Click LCA's Carbon Footprint (One-Click LCA® software, n.d.).

The closest early-stage calculator is Cardinal LCA, a plugin for Grasshopper3D based in Rhino3D. It focuses on stages A1-A3, which, similar to LearnCarbon, includes product stages. However, even though it has the flexibility of using custom data, it does not focus on the geometry exploration phase of design. It requires a developed 2.5D or 3D model for EE calculation.

EMBODIED CARBON BENCHMARK STUDY

LearnCarbon uses ML for the two prediction use cases. We used The Embodied Carbon Benchmark Study (Carbon Leadership Forum, 2020, Simonen et al., 2017) as a dataset for training. The study contains benchmarks for GWPwith respect to parameters of Location, Building Type (BT), Floor Count (FC), and Gross Floor Area (GFA).

This dataset was used due to several advantages, like the environmental benchmarks for buildings and a technical report by the Joint Research Centre (JRC) (Gervasio and Dimova, n.d.). The data was open-source and available in an editable format. Further, it reflected the variation in EE between different building types and scales, which provides rich feedback. In addition, the data is supported by leading engineering firms along with the International Living Future Institute, The DeQo (De Wolf et al. 2016) Database/MIT, the WRAP Database, and data compiled by the UW team.

The study has certain limitations, some of which are identified in the report but are re-iterated to validate how it impacts training the models. This database is not a statistically representative sample of current building practices and is weighted to larger, more prominent buildings than those that make up the complete building stock. Comparing different typologies and scales is not directly

proportional, and there are uncertainties in the process. Emphasis is laid only on the GWP. However, in various building typologies like high-rise commercial buildings, envelopes will considerably contribute to the total lifetime emissions (Eurostat, 2017). There is no indication of the structural typology, which is an essential aspect while using the database for other use cases.

ML AND ENVIRONMENTAL IMPACT ASSESSMENTS

LCA and other Environmental Impact Assessment (EIA) frameworks are increasingly being used (Means and Guggemos, 2015). Such assessment requires the encoding and detailed enumeration of all building components, which may result in insufficient or false results due to information reduction. To counterbalance that, D'Amico et al. (2019) suggest combining existing EIA methods with ML techniques.

Research on ML applied to structural engineering has been nothing new since it started in 1989 (Adeli and Yeh, 2008). Most modern applications focus on the prediction of buckling loads (Mukherjee, Deshpande, and Anmala, 1996; Ly et al., 2019), material elastic properties (Al Abadi et al., 2018), and the refining finite element models (Carrera, Kaleel and Petrolo, 2017). However, there is, to our knowledge, no ML application in predicting EE with structural material, a gap that LearnCarbon attempts to address. According to D'Amico et al. (2019), the lack of such prediction models is caused by the inexistence of datasets competent enough for training.

A Non-linear Regression model

Nonlinear regression models are statistical models for understanding complex relations in data. Mendecka et al. (2019) used nonlinear regression to predict GWP along with other impact metrics for the wind energy industry. Additionally, methods for EIA of an industrial product under different operating conditions were formulated using the same ML tool (Lozano, Cabrera, and Blanco-Marigorta, 2020).

A Classification model

Classification predictive modeling involves assigning a class label describing an attribute to input data. In LearnCarbon, we utilize the multi-class classification, which consists in predicting one of more than two classes for input data. There are several examples of classification algorithms in the field. Wang et al. use a classification model to predict construction cost and schedule success (Wang, Yu, and Chan, 2012) before the building's realization. Other frameworks aim to classify semi- rigid frames (Goto and Miyashita, 1998) or distinguish shear-dominated from flexure-dominated columns (Zhu, Elwood, and Haukaas, n.d.).

METHODOLOGY

Foreword to the methodology used to realize LearnCarbon, a brief description of the data flow is demonstrated (supported by Figure 1). Initially, the dataset is processed for training regression and a classification model. The plugin calls these two models to predict the GWP and the construction type, respectively. LearnCarbon's interface allows flexibility in setting the parameters and presents the models' predictions to the user.

Construction Type Framework

Initially, we lacked information on the material type and quantities of each construction type, see section ***Embodied Carbon Benchmark Study***. Therefore, to simplify the dataset and assign material amounts and types and reduce the possible variation in structural types, we categorized the various buildings into four distinct construction types: Reinforced Concrete, Steel and Concrete Hybrid, Timber, and Timber Hybrid (Ecohome, 2017). This was followed by assigning a structural category for each building based on heuristic assumptions, see Table 1.

The above assumptions were used for a pilot project for the hackathon. For future development, these assumptions should be further refined for accurate results. These assumptions were

satisfactory for the proof of concept to move forward with the ML model downstream.

Data Processing

The dataset consists of categorical and numerical features which describe building projects. However, some categorical data were converted into numerical with pandas (Mckinney, 2010) and NumPy (Harris et al., 2020) libraries. This included the GFA (93m or less, 94m to 465m, etc.) and FC (1 to 6, 7 to 14, etc.) and was converted into real value numbers based on the GWP emissions. The equivalent range in GWPwas calculated for each category, so a value can be assigned for each entry with linear interpolation. For example, a building with a high GWPvalue in the equivalent range of the category 1 to 6 number of floors will have 5 or 6 feet.

The purpose of this conversion refers back to our initial aim to predict a GWPvalue based on a massing model and construction type in the early design stage. While Rhino3D allows us to extract information on FC and GFA with a known Floor-To-Floor height (FTF), the selected features did not match the requirements to achieve prediction. The method chosen to fill data gaps may result in inaccurate predictions. As outlined in future work, we aim to re-train the model with a detailed database. Renovation projects were limited in the database; therefore, only new construction projects were included. BT was described by Commercial and Residential, excluding Generic and Industrial, while different continents described the Location.

The processing approach offered different possibilities and limitations. While developing the framework, questions on correlations between separate entries and the possibility of training an ML model emerged. Among other indications in Figure 2, we observe that the features of Building Construction Type (BCT) and GWP are negatively correlated, which showcases the potential of teaching this to a model. Another relevant indication to our hypothesis is between FC with Location and GFA with CO2. While the latter logically form a strong relationship, the former points out a potential bias in the dataset or even an observation of building heights per continent.

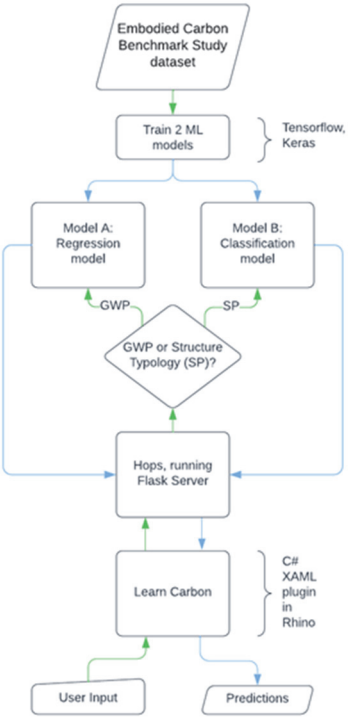
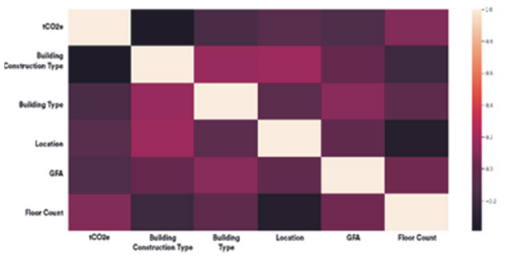


Figure 1
The Data Flow Diagram of LearnCarbon. The green arrows indicate the flow of input data, while the blue arrows indicate the output data flow.

If GWP>300	If 1<Floor Count<6	Reinforced Concrete
If GWP<300	If 1<Floor Count<6	Timber
If GWP>1000	If Floor Count>25	Steel + Reinforced Concrete
If GWP<1000	If Floor Count>25	Reinforced Concrete
If GWP>250	If 1<Floor Count<6	Timber Hybrid
if GWP>300	If 7<Floor Count<14	Reinforced Concrete
if GWP<300	If 7<Floor Count<14	Timber Hybrid

Table 1
Assumptions for construction types as compared to their GWP measured in tons of C02 equivalent (tC02e).

Figure 2
Correlation matrix of the features



Network Architecture and Training

The features after data processing were used for the supervised learning process developed using Keras (Keras-team, 2019) from Tensorflow (TensorFlow, 2019). The hypothesis was explored based on two scenarios, which, as in the work of Quinteros et al. (REF), were developed through a back-propagation algorithm training a feed-forward Neural Network (NN). The first scenario is achieved with a non-linear regression model trained to predict a continuous value, GWP, based on five features: BT, Location, GFA, FC, and BCT. Before passing the data to the model, the input features were scaled with a Standard Scaler and the output features with a MinMax Scaler to match the distribution in the data. The model is structured with three dense layers of 50, 32, and 1 neuron respectively. The first two consist of the hidden layers, which were activated with the Relu function, during the last one with the Sigmoid function. The model was compiled with the loss function of Mean-Squared-Error (MSE) and with the use of the Adam optimizer.

The second scenario is achieved by training a classification model which predicts the construction type based on the GWP, BT, location, GFA, and floor count. It is structured with input and four dense layers of 128, 64, 32, and 4 neurons, respectively. The first three hidden layers were activated using the Relu function, while the last one, representing the output, was activated with the softmax function. Both models were trained for 200 epochs. The first scenario seemed to reach satisfactory results faster, while the last one started showing signs of overfitting.

Connectivity

After training, the next step was connecting with Rhino3d. We used the Hops (Food4Rhino, 2021b) component in Rhino3d that allowed for integrating Tensorflow and Numpy into Rhino 3d CPython. The Hops infrastructure developed by McNeel is a thin wrapper around an underlying Flask server that allows running a flask server and serving data from an external python instance. Therefore, the pipeline was free to call all the necessary dependencies outside the Rhino instance.

The client-facing UI (Figure 3) was built in C# with XAML, allowing user input. After designing a 3D massing sketch, the user can proceed by selecting and defining inputs: Location, BT, and FTF. The last one is used to calculate the GFA based on the height of the building, as well as the footprint. The user is also called to select the type of prediction. BCT is required for the first scenario, BCT is required for the second, the target tCO2e.

Once the data is passed, the Flask server (KPF Urban Interface, 2022) automatically detects the changes made to the inputs and re-runs the code, which loads the ML model with the newly indicated inputs. The prediction is passed through Hops to the UI in Rhino3D. Notably, due to Flask Server running locally, data is served instantly.

MODEL EVALUATION

The evaluation process of both models was conducted by testing unseen entries from the same data set. The dataset was split into 582 entries for training and 146 for testing, 80% and 20%, respectively. The NN from the first scenario achieved a loss of 0.0016, which refers to the MSE selection that was chosen while compiling the model. The output number indicates a close fit to the ground truth data (Figure 4), and the model was able to learn the relationship between the input and the output up to a satisfactory level for this context of the early design phase.

The classification network for the second scenario reached validation accuracy levels of 94%

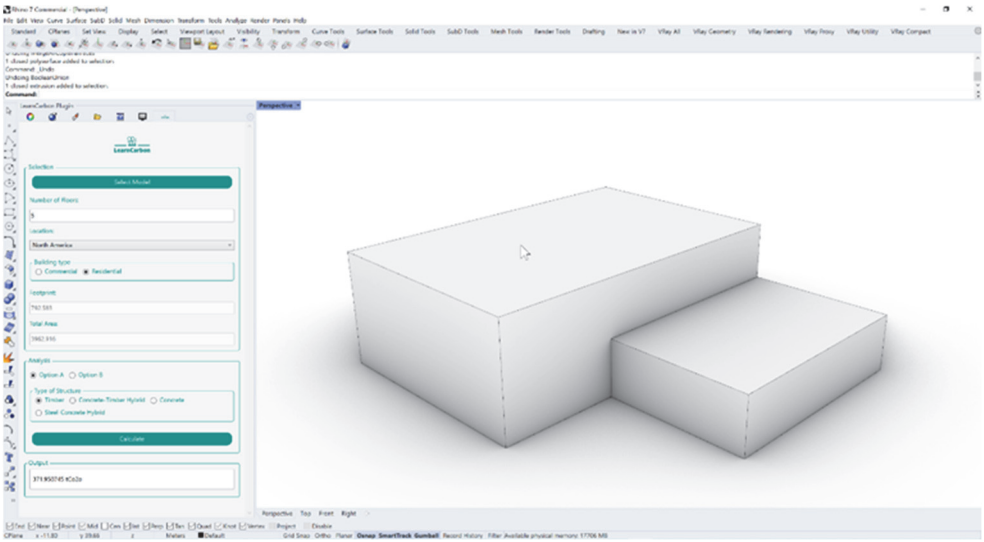


Figure 3
LeanCarbon's interface

on the test data. The diagonal squares in Figure 5 show the successful predictions from the test data. However, some predictions didn't match the ground truth, such as the 3 entries classified as Steel-Concrete instead of Concrete. As with all ML models, even though the algorithms used for training operate in a stochastic domain using randomness for the initialization, the trained models are deterministic systems. The same input will always result in the same output.

LEARN CARBON RESULTS

To study the tool's validity, a study comparing results from LearnCarbon with the EC3 online tool was performed (A1-A3). Three sketches were developed for residential typology (BT) located in North America (Location), having a reinforced concrete structure (BCT). In EC3, the selected settings were 100 Concept Design (Level of Development), New Construction Project, Concrete PT Framing as the Primary Horizontal Gravity System, Concrete CIP as the primary Vertical Gravity System, and Concrete Shear Walls as the primary lateral Resistance System.

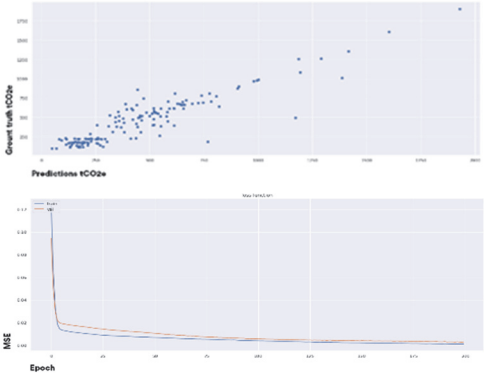


Figure 4
Top: A scatter plot of GWP versus Ground truth values.
Bottom: A plot of the Mean Squared Error function throughout the epochs.

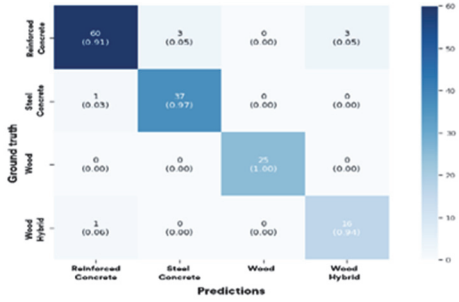
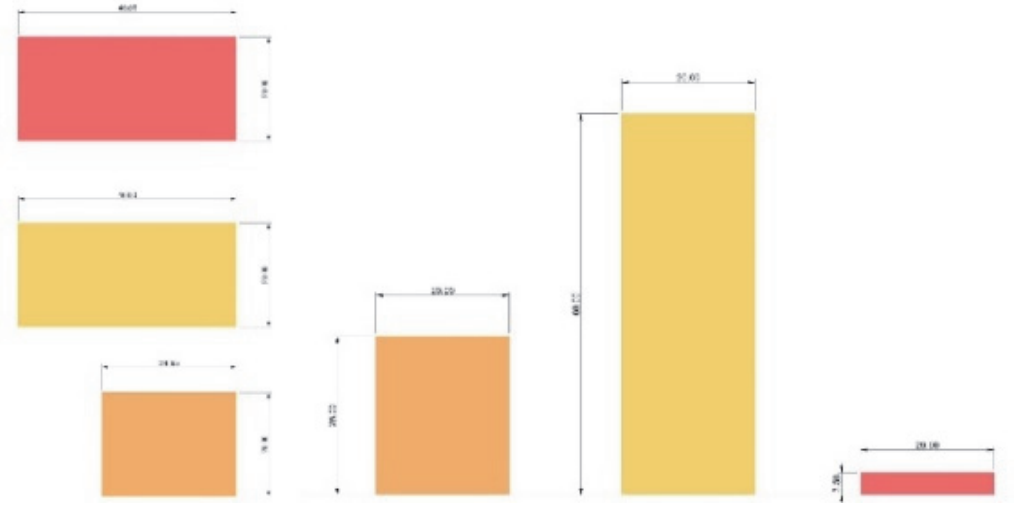


Figure 5
The plot of Structural typology Predictions versus ground truth.

Table 2
A comparison between the predictions of LearnCarbon and the assumptions of EC3.

Building	LearnCarbon	EC3
Red (40x20x3.5m)	541 tCO2e	382 tCO2e
Orange (24.6x20x20m)	1125 tCO2e	1163 tCO2e
Yellow (40x20x60m)	206 tCO2e	3565 tCO2e

Figure 6
Geometries for comparing the accuracy of Learn Carbon in top view and front-view (Rhino3D). Three geometries were sketched with the following properties and studied with Learn Carbon and EC3.



In all three buildings, concrete average values were considered for structure, including the core. In LearnCarbon, similar BT, BCT, and Location settings were used as the input. The results between the two frameworks can be seen in Table 2 and Figure 6.

The test highlights that the values from both the tools are comparable, except for the third case, where the values from LearnCarbon are shallow. This is attributed to training data having fewer entries for high-rise buildings.

DISCUSSION AND FUTURE PERSPECTIVES

LearnCarbon aims to inform the design regarding environmental impacts by predicting the GWP of buildings early on. As stated in the Background section, there is a gap in the available tools for the particular but impactful project phase. Additionally, the tool can assist the field in making sustainable decisions by predicting construction types for future designs. LearnCarbon serves as a proof of concept for implementing ML in EIA frameworks. ML is an appropriate methodology in this case because it recognizes building patterns from a dataset of multiple entries and correlated features in comparison with statistical models.

LearnCarbon is, at its current state, successfully predicting GWP values based on the aforementioned input data, while the models' performance is sufficient with a loss function of 0.0016 in model A and 94% accuracy in model B, see section ***Model Evaluation***. As mentioned in the section ***Embodied Carbon Benchmark Study***, the dataset that we used to train our models contains phase A for calculating the GWP of a building. In this case, the models are trained to understand the relationship between their GWP, the construction type, the size of the building (GFA, FC), and its location. Therefore, the choice of parameters for the aforementioned categories alters the results and can provide insights regarding the environmental impact.

However, the tool fails in predicting accurate results for extreme values since the dataset that trained the models had few or zero data points describing such examples. Additional biases are implanted in the model regarding the chronological period, construction frameworks, and operational methods of buildings. New questions emerge regarding the implementation of other methods for lower. To counterbalance that, the models must be retrained with more recent datasets and reevaluated. Additionally, for the creation of contemporary and new resource-aware models, we urge the field, both industry, and academia, to share their data publicly for better results. Accurate material quantities, including percentages for structural materials, could also offer more specific results. Apart from the backend code, which is software agnostic, the front-end is constructed for Rhino3D, an environment that could host an interface to wrap the computation. In future work, through a cloud connection, a simple API could allow one to request predictions from a publically available endpoint and process it in an environment of choice.

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